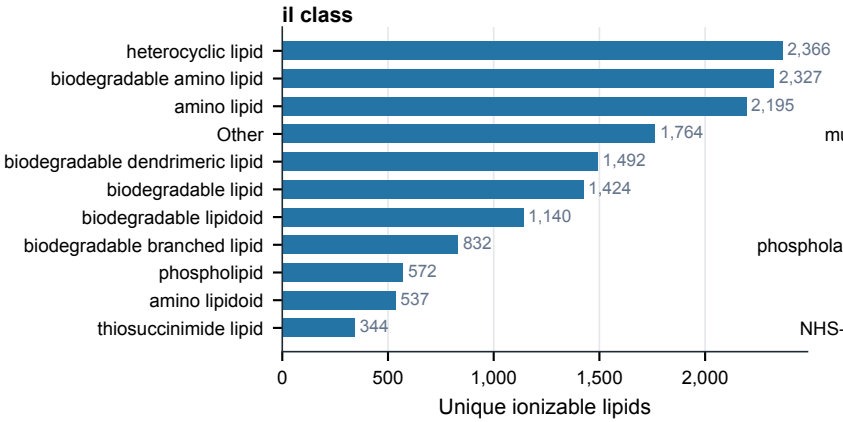
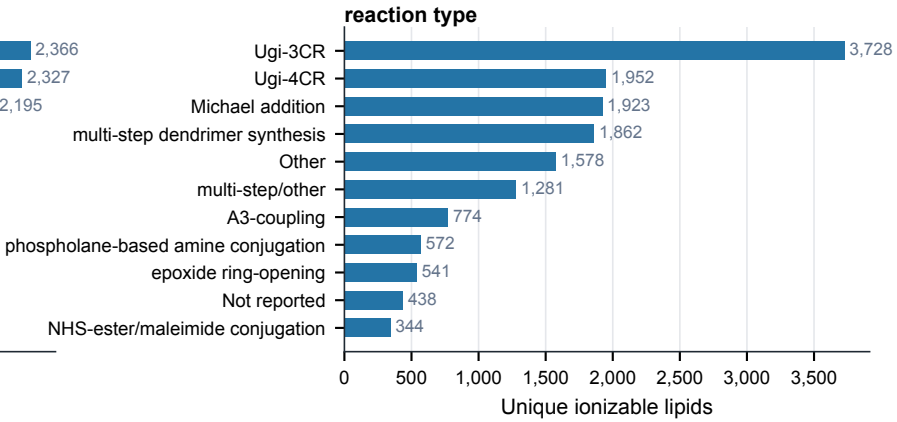
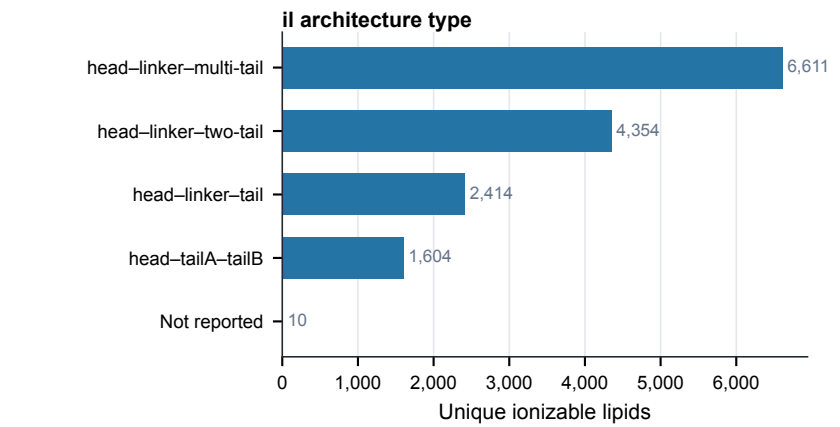
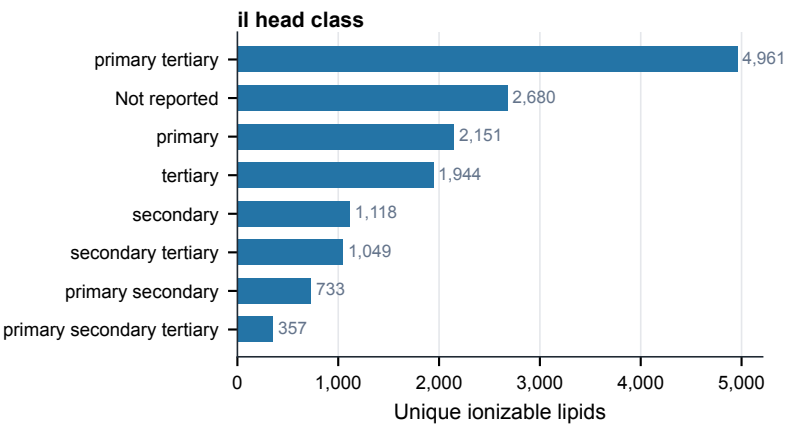
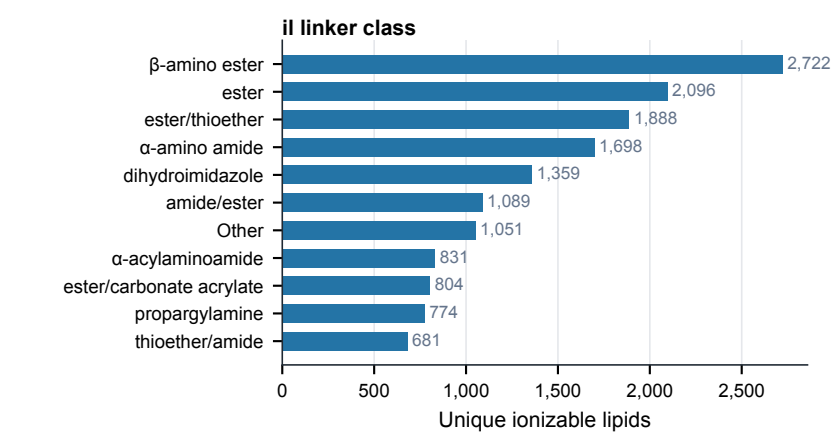
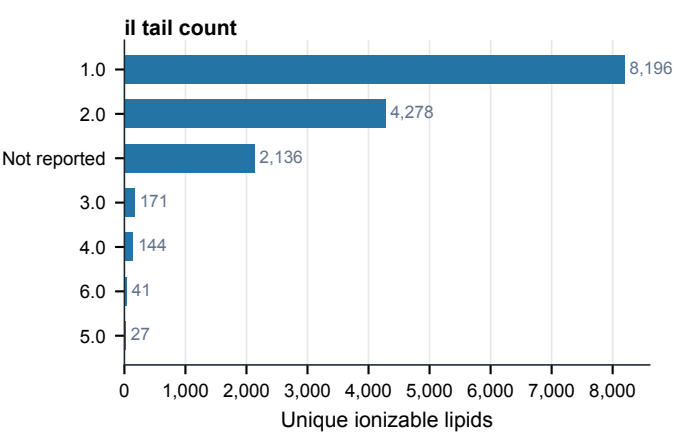

il class
heterocyclic lipid
biodegradable amino lipid
amino lipid
Other
biodegradable dendrimeric lipid
biodegradable lipid
biodegradable lipidoid
biodegradable branched lipid
phospholipid
amino lipidoid
thiosuccinimide lipid
2,366
2,327
2,195
1,764
1,492
1,424
1,140
832
572
537
344
0
500
1,000
1,500
2,000
Unique ionizable lipids


reaction type
Ugi-3CR
Ugi-4CR
Michael addition
multi-step dendrimer synthesis
Other
multi-step/other
A3-coupling
phospholane-based amine conjugation
epoxide ring-opening
Not reported
NHS-ester/maleimide conjugation
3,728
1,952
1,923
1,862
1,578
1,281
774
572
541
438
344
0
500
1,000
1,500
2,000
2,500
3,000
3,500
Unique ionizable lipids


il architecture type
head–linker–multi-tail
head–linker–two-tail
head–linker–tail
head–tailA–tailB
Not reported
6,611
4,354
2,414
1,604
10
0
1,000
2,000
3,000
4,000
5,000
6,000
Unique ionizable lipids


il head class
primary tertiary
Not reported
primary
tertiary
secondary
secondary tertiary
primary secondary
primary secondary tertiary
4,961
2,680
2,151
1,944
1,118
1,049
733
357
0
1,000
2,000
3,000
4,000
5,000
Unique ionizable lipids


il linker class
β-amino ester
ester
ester/thioether
α-amino amide
dihydroimidazole
amide/ester
Other
α-acylaminoamide
ester/carbonate acrylate
propargylamine
thioether/amide
2,722
2,096
1,888
1,698
1,359
1,089
1,051
831
804
774
681
0
500
1,000
1,500
2,000
2,500
Unique ionizable lipids


il tail count
1.0
2.0
Not reported
3.0
4.0
6.0
5.0
8,196
4,278
2,136
171
144
41
27
0
1,000
2,000
3,000
4,000
5,000
6,000
7,000
8,000
Unique ionizable lipids